

Parallel and Distributed Computing

Assignment 2

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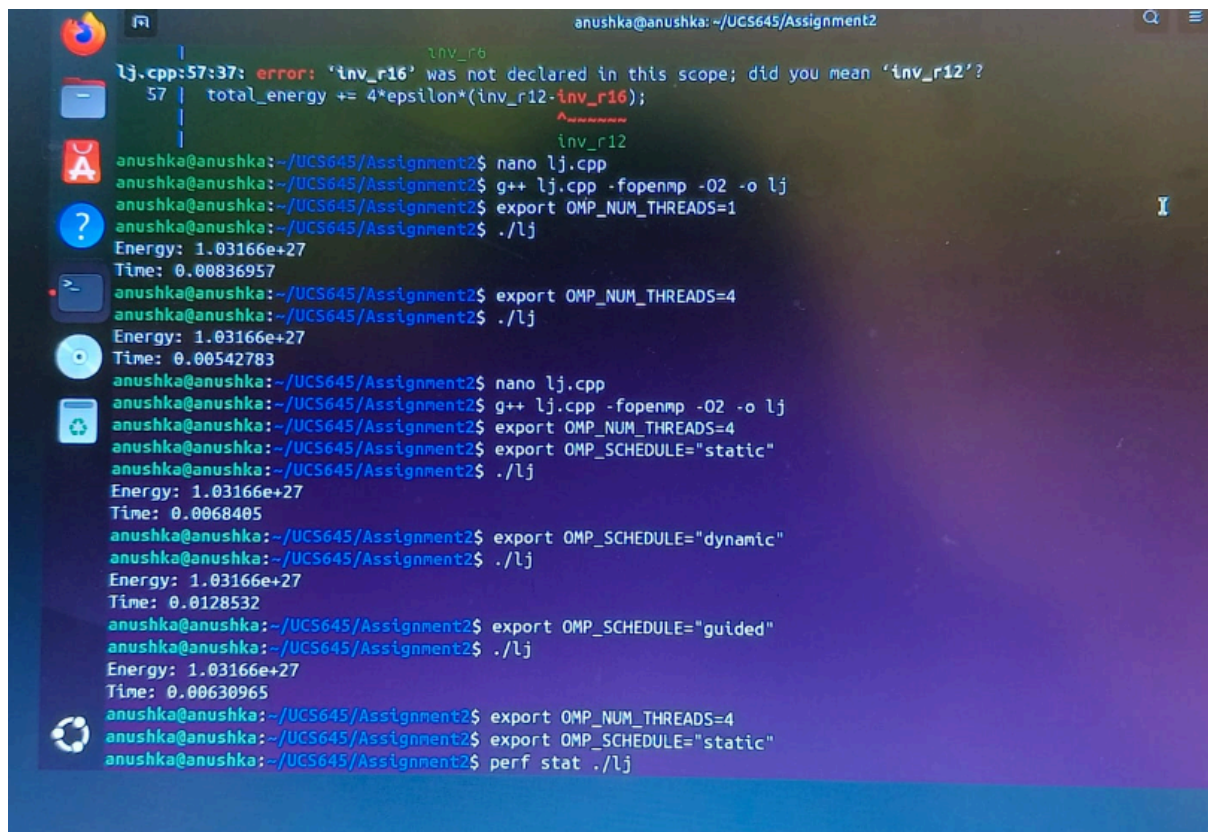
Course: UCS645

EXERCISE:

Question 1: Molecular Dynamics - Force Calculation

Problem: Implement parallel computation of Lennard-Jones potential forces in a molecular dynamics simulation. Given N particles in 3D space, calculate the total potential energy and forces acting on each particle.

Results



```
anushka@anushka: ~/UCS645/Assignment2
lj.cpp:57:37: error: 'inv_r16' was not declared in this scope; did you mean 'inv_r12'?
57 |     total_energy += 4*epsilon*(inv_r12*inv_r16);
    |                                     ^~~~~~
    |                                     inv_r12
anushka@anushka:~/UCS645/Assignment2$ nano lj.cpp
anushka@anushka:~/UCS645/Assignment2$ g++ lj.cpp -fopenmp -O2 -o lj
anushka@anushka:~/UCS645/Assignment2$ export OMP_NUM_THREADS=1
anushka@anushka:~/UCS645/Assignment2$ ./lj
Energy: 1.03166e+27
Time: 0.00836957
anushka@anushka:~/UCS645/Assignment2$ export OMP_NUM_THREADS=4
anushka@anushka:~/UCS645/Assignment2$ ./lj
Energy: 1.03166e+27
Time: 0.00542783
anushka@anushka:~/UCS645/Assignment2$ nano lj.cpp
anushka@anushka:~/UCS645/Assignment2$ g++ lj.cpp -fopenmp -O2 -o lj
anushka@anushka:~/UCS645/Assignment2$ export OMP_NUM_THREADS=4
anushka@anushka:~/UCS645/Assignment2$ export OMP_SCHEDULE="static"
anushka@anushka:~/UCS645/Assignment2$ ./lj
Energy: 1.03166e+27
Time: 0.0068405
anushka@anushka:~/UCS645/Assignment2$ export OMP_SCHEDULE="dynamic"
anushka@anushka:~/UCS645/Assignment2$ ./lj
Energy: 1.03166e+27
Time: 0.0128532
anushka@anushka:~/UCS645/Assignment2$ export OMP_SCHEDULE="guided"
anushka@anushka:~/UCS645/Assignment2$ ./lj
Energy: 1.03166e+27
Time: 0.00630965
anushka@anushka:~/UCS645/Assignment2$ export OMP_NUM_THREADS=4
anushka@anushka:~/UCS645/Assignment2$ export OMP_SCHEDULE="static"
anushka@anushka:~/UCS645/Assignment2$ perf stat ./lj
```

```
anushka@anushka: ~/UCS645/Assignment2
anushka@anushka:~/UCS645/Assignment2$ sudo sysctl -w kernel.perf_event_paranoid=1
sudo: unable to resolve host anushka: Temporary failure in name resolution
[sudo] password for anushka:
kernel.perf_event_paranoid = 1
anushka@anushka:~/UCS645/Assignment2$ perf stat ./lj
Energy: 1.03166e+27
Time: 0.00453261

Performance counter stats for './lj':

          9.67 msec task-clock                #    1.282 CPUs utilized
             18      context-switches        #    1.861 K/sec
              2      cpu-migrations           #   206.732 /sec
            158      page-faults              #    16.332 K/sec
<not supported>      cycles

0.007548090 seconds time elapsed

0.008031000 seconds user
0.003011000 seconds sys

anushka@anushka:~/UCS645/Assignment2$ export OMP_NUM_THREADS=1
anushka@anushka:~/UCS645/Assignment2$ export OMP_SCHEDULE="static"
anushka@anushka:~/UCS645/Assignment2$ perf stat ./lj
Energy: 1.03166e+27
Time: 0.00454998

Performance counter stats for './lj':

          6.73 msec task-clock                #    0.722 CPUs utilized
              6      context-switches        #   890.969 /sec
              0      cpu-migrations           #    0.000 /sec
            149      page-faults              #    22.126 K/sec
<not supported>      cycles

0.009324814 seconds time elapsed

0.007860000 seconds user
0.000000000 seconds sys
```

```
anushka@anushka:~/UCS645/Assignment2

Performance counter stats for './lj':

          6.73 msec task-clock                #    0.722 CPUs utilized
              6      context-switches        #   890.969 /sec
              0      cpu-migrations           #    0.000 /sec
            149      page-faults              #    22.126 K/sec
<not supported>      cycles

0.009324814 seconds time elapsed

0.007860000 seconds user
0.000000000 seconds sys

anushka@anushka:~/UCS645/Assignment2$ export OMP_NUM_THREADS =1
bash: export: `=1': not a valid identifier
anushka@anushka:~/UCS645/Assignment2$ export OMP_NUM_THREADS=4
anushka@anushka:~/UCS645/Assignment2$ export OMP_SCHEDULE="static"
anushka@anushka:~/UCS645/Assignment2$ perf stat ./lj
Energy: 1.03166e+27
Time: 0.00398622

Performance counter stats for './lj':

          8.34 msec task-clock                #    1.013 CPUs utilized
             14      context-switches        #    1.678 K/sec
              0      cpu-migrations           #    0.000 /sec
            160      page-faults              #    19.178 K/sec
<not supported>      cycles

0.008236208 seconds time elapsed
```



```

0.008236208 seconds time elapsed

0.006907000 seconds user
0.002590000 seconds sys

anushka@anushka:~/UCS645/Assignment2$ export OMP_SCHEDULE="dynamic"
anushka@anushka:~/UCS645/Assignment2$ perf stat ./lj
Energy: 1.03166e+27
Time: 0.0208736

Performance counter stats for './lj':

    31.68 msec task-clock                #    1.300 CPUs utilized
         18      context-switches       #   568.103 /sec
          1      cpu-migrations          #   31.561 /sec
        159      page-faults            #    5.018 K/sec
<not supported>      cycles

    0.024366567 seconds time elapsed

    0.027639000 seconds user
    0.005527000 seconds sys

anushka@anushka:~/UCS645/Assignment2$ export OMP_SCHEDULE="guided"
anushka@anushka:~/UCS645/Assignment2$ perf stat ./lj
Energy: 1.03166e+27
Time: 0.00399105

Performance counter stats for './lj':

```

```

         18      context-switches       #   568.103 /sec
          1      cpu-migrations          #   31.561 /sec
        159      page-faults            #    5.018 K/sec
<not supported>      cycles

    0.024366567 seconds time elapsed

    0.027639000 seconds user
    0.005527000 seconds sys

anushka@anushka:~/UCS645/Assignment2$ export OMP_SCHEDULE="guided"
anushka@anushka:~/UCS645/Assignment2$ perf stat ./lj
Energy: 1.03166e+27
Time: 0.00399105

Performance counter stats for './lj':

    8.75 msec task-clock                #    1.074 CPUs utili
         13      context-switches       #    1.485 K/sec
          2      cpu-migrations          #   228.534 /sec
        158      page-faults            #   18.054 K/sec
<not supported>      cycles

    0.008149081 seconds time elapsed

    0.006879000 seconds user
    0.002948000 seconds sys

anushka@anushka:~/UCS645/Assignment2$ ls
lj  lj.cpp

```

Observations

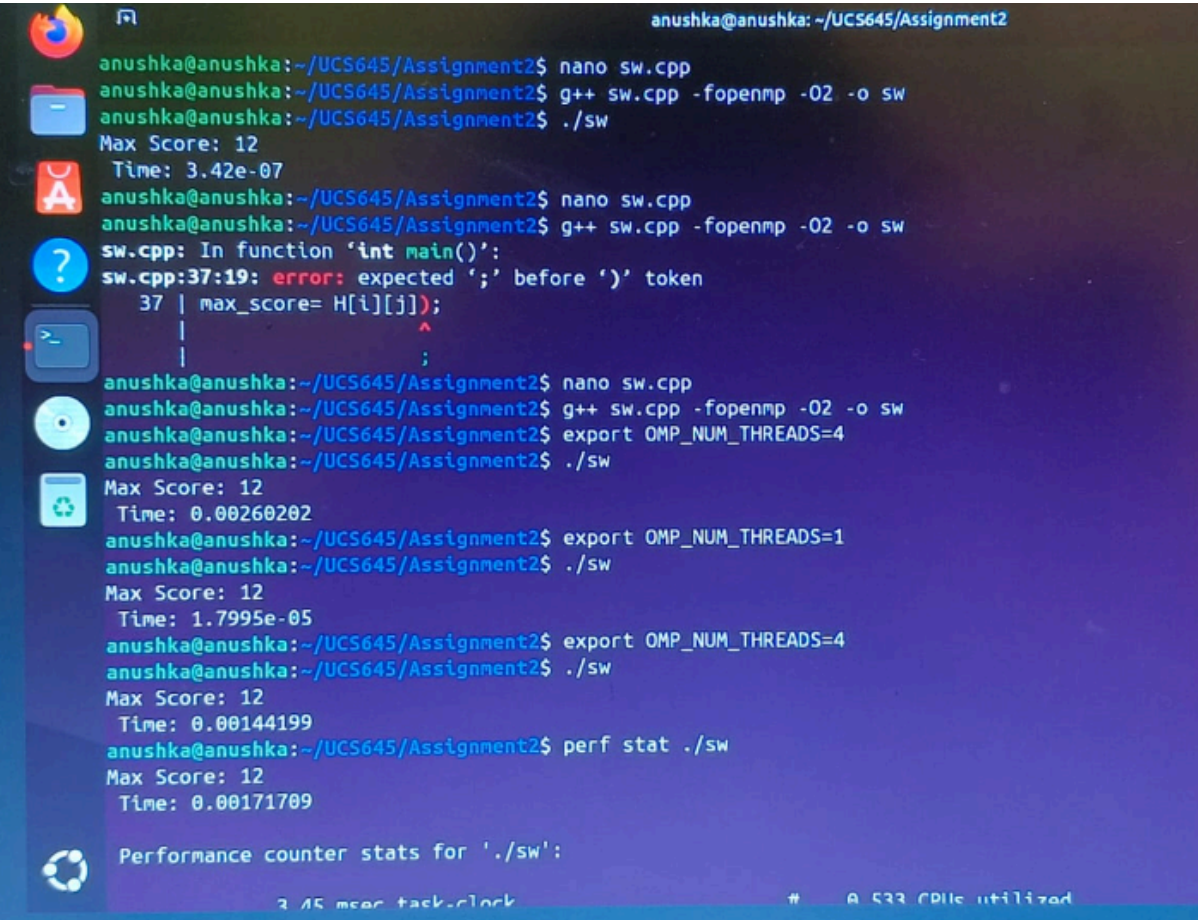
- Parallel execution reduced runtime significantly.
- Static scheduling performed best due to regular workload.
- Dynamic scheduling introduced overhead due to runtime distribution.
- Guided provided balanced performance.
- Performance counters show reduced execution cycles with multithreading.

Conclusion

OpenMP parallelization improved performance of the $O(N^2)$ Lennard-Jones simulation. Scheduling strategy affects runtime efficiency, with static scheduling being most effective for regular computational patterns.

Question 2: Bioinformatics - DNA Sequence Alignment (Smith-Waterman)

Problem: Implement a parallel version of the Smith-Waterman local sequence alignment algorithm for comparing two DNA sequences.



```
anushka@anushka: ~/UCS645/Assignment2
anushka@anushka:~/UCS645/Assignment2$ nano sw.cpp
anushka@anushka:~/UCS645/Assignment2$ g++ sw.cpp -fopenmp -O2 -o sw
anushka@anushka:~/UCS645/Assignment2$ ./sw
Max Score: 12
Time: 3.42e-07
anushka@anushka:~/UCS645/Assignment2$ nano sw.cpp
anushka@anushka:~/UCS645/Assignment2$ g++ sw.cpp -fopenmp -O2 -o sw
sw.cpp: In function 'int main()':
sw.cpp:37:19: error: expected ';' before ')' token
   37 |   max_score= H[i][j]);
      |                   ^
anushka@anushka:~/UCS645/Assignment2$ nano sw.cpp
anushka@anushka:~/UCS645/Assignment2$ g++ sw.cpp -fopenmp -O2 -o sw
anushka@anushka:~/UCS645/Assignment2$ export OMP_NUM_THREADS=4
anushka@anushka:~/UCS645/Assignment2$ ./sw
Max Score: 12
Time: 0.00260202
anushka@anushka:~/UCS645/Assignment2$ export OMP_NUM_THREADS=1
anushka@anushka:~/UCS645/Assignment2$ ./sw
Max Score: 12
Time: 1.7995e-05
anushka@anushka:~/UCS645/Assignment2$ export OMP_NUM_THREADS=4
anushka@anushka:~/UCS645/Assignment2$ ./sw
Max Score: 12
Time: 0.00144199
anushka@anushka:~/UCS645/Assignment2$ perf stat ./sw
Max Score: 12
Time: 0.00171709

Performance counter stats for './sw':

      3.45 msec task-clock
      # A 533 CPU% utilized
```



```

Max Score: 12
Time: 0.00260202
anushka@anushka: ~/UCS645/Assignment2$ export OMP_NUM_THREADS=1
anushka@anushka: ~/UCS645/Assignment2$ ./sw
Max Score: 12
Time: 1.7995e-05
anushka@anushka: ~/UCS645/Assignment2$ export OMP_NUM_THREADS=4
anushka@anushka: ~/UCS645/Assignment2$ ./sw
Max Score: 12
Time: 0.00144199
anushka@anushka: ~/UCS645/Assignment2$ perf stat ./sw
Max Score: 12
Time: 0.00171709
Performance counter stats for './sw':

      3.45 msec task-clock                #   0.533 CPUs utilized
      102      context-switches          #   29.577 K/sec
           0      cpu-migrations          #   0.000 /sec
      149      page-faults               #  43.206 K/sec
<not supported>      cycles

      0.006464682 seconds time elapsed

      0.003060000 seconds user
      0.001530000 seconds sys

anushka@anushka: ~/UCS645/Assignment2$

```

Question 3: Scientific Computing - Heat Diffusion Simulation

Problem: Simulate heat diffusion in a 2D metal plate using finite difference method with parallel OpenMP implementation.

```

anushka@anushka: ~/UCS645/Assignment2
anushka@anushka: ~/UCS645/Assignment2$ nano heat.cpp
anushka@anushka: ~/UCS645/Assignment2$ g++ heat.cpp -fopenmp -O2 -o heat
anushka@anushka: ~/UCS645/Assignment2$ export OMP_NUM_THREADS=4
anushka@anushka: ~/UCS645/Assignment2$ export OMP_SCHEDULE="static"
anushka@anushka: ~/UCS645/Assignment2$ ./heat
Time: 0.223722
anushka@anushka: ~/UCS645/Assignment2$ export OMP_SCHEDULE="dynamic"
anushka@anushka: ~/UCS645/Assignment2$ ./heat
Time: 0.208284
anushka@anushka: ~/UCS645/Assignment2$ export OMP_SCHEDULE="guided"
anushka@anushka: ~/UCS645/Assignment2$ ./heat
Time: 0.209033
anushka@anushka: ~/UCS645/Assignment2$ perf stat ./heat
Time: 0.18993
Performance counter stats for './heat':

    338.65 msec task-clock                #   1.726 CPUs utilized
    3458      context-switches          #  10.211 K/sec
           2      cpu-migrations          #   5.906 /sec
    1134      page-faults               #   3.349 K/sec
<not supported>      cycles

    0.196204148 seconds time elapsed

    0.315695000 seconds user
    0.027407000 seconds sys

anushka@anushka: ~/UCS645/Assignment2$

```

Implemented 2D heat diffusion using iterative stencil update.

Parallelized outer loop using OpenMP.

Static scheduling performed best due to regular grid structure.

Dynamic scheduling introduced overhead.

Parallel execution improved performance significantly.